

6

Child second language, multilingual and heritage language acquisition, language attrition

6.1 Adult L2 acquisition	142	6.6 Commonalities and differences	
6.2 Child L2 acquisition	142	between the four acquisition	
6.3 L3/Ln acquisition	147	contexts	161
6.4 Heritage language learners	153	6.7 Exercises	168
6.5 Language attrition	158		

Starting in the previous chapter and in this one, we are thinking about individuals with two or more languages in their brain. There is one issue that unites all bilinguals and multilinguals, no matter when the other languages were learned: their minds/brains are not identical in their functionality to those of monolinguals, precisely because the former operate with two and more languages. This fact is of paramount importance when discussing multilingual linguistic behavior. It may confer a broader cognitive benefit, but in day-to-day language use, it may also make lexical access harder and processing more taxing.

In this chapter, I will look at various language acquisition and change conditions. I will briefly introduce the process of adult L2 acquisition, which is the focus of this book. Next, I compare child and adult L2, to see

if these are qualitatively different processes. I will then present L3 or L_n language, or multilingual, acquisition. Heritage language speakers are bilingual speakers who are using their native language while speaking another, dominant language. Language attrition, the loss of the native language, may arise under the conditions of bilingualism even if speakers are late bilinguals. All of these conditions of incrementally adding or losing a language will help us think about the important factors affecting these changes. This chapter paints a multifactorial picture of language acquisition. At the end of this chapter, you will be able to appreciate some of the important factors and variables that researchers take into account in order to investigate the process. But first, let me introduce adult native speakers.

6.1 Adult L2 acquisition

Who can be considered an adult L2 acquirer? The current cutoff point for considering someone an adult L2 learner is 7 or 8 years of age, based on Johnson and Newport's classical (1989, 1991) studies. However, there has been some variation in this respect in the literature. For example, Lenneberg proposed 6 years of age to be the end of the Critical Period for language acquisition. Another commonly used cutoff point is puberty (between 11 and 14 years of age nowadays). As the rest of this textbook deals with the way adults acquire a second language, we will not spend too much time in this section describing acquisition routes. However, remember that the conditions of acquisition (in the country where the language is spoken or not), type of exposure (naturalistic or classroom), and length of language exposure and use may matter at least as much as age for the convergence of the speakers on a nativelike mental grammar.

6.2 Child L2 acquisition

When the *initial* exposure to the nonnative language is approximately between the ages of 4 and 7, we define it as child L2 (cL2) acquisition (Schwartz 2004, Meisel 2011). The reasoning for the lower limit is that 4 is considered the age when the bulk of the native grammar is roughly in

place.¹ The age of 7 or 8 has been argued to be the critical age after which acquisition may not proceed as in child L1. Although opinions in the literature vary, there is some experimental support for these cut off ages. For example, the classic studies by Johnson and Newport (1989, 1991) identified children who started to acquire the L2 before this age as most likely to fall in native ranges of competence.

My approach here will follow Bonnie Schwartz's groundbreaking work on this topic (Schwartz 1992, 2004, 2009, see also Lakshmanan 1995). She calls the child–adult L2 comparison “the perfect natural experiment” (Schwartz 2004: 17). Why would that be? What is the rationale for considering L2 children such an important population, with respect to the UG issue? Recall from Chapter 4 that, according to the Critical Period Hypothesis, second language acquisition cannot proceed as child language acquisition, if the learner is already an adult (aL2). This logic can be extended as follows: if the learner is not an adult, that is, between the ages of 4 and 7 or 8, L2 acquisition should proceed differently. For example, in Johnson and Newport's (1989) study, the child acquirers were generally more successful than those who started as adults, and a negative age-to-nativeness correlation was observed. Thus, if research uncovers evidence that the developmental *paths* of L2 children and adults look the same, in the sense that they make similar errors and exhibit the same developmental stages, this would constitute evidence that they are both going the UG-sanctioned route. If, on the other hand, the developmental course of the adult L2 learner diverges from that of the child L2 learner, this would be evidence that adult L2 acquisition is not UG-constrained and may be due to rote learning, various problem-solving strategies, and superficial noticing of linguistic rules. The L1–cL2–aL2 comparison, then, can address a seminal research question, namely, the question that we have been pondering in this textbook from the very beginning.

What are some similarities and differences between the three populations we are considering here? Although the latter may be more mature, L1 and cL2 learners are children, that is, they fall within the window of opportunity of the Critical Period for language. On the other hand, cL2 and aL2 learners are similar in that they have a previously acquired language, their native language. In contemplating whether the same underlying processes are

¹ Although we saw in the previous chapter that a lot of complex constructions are not acquired by children until much later than the age of 4.

involved in L1, cL2, and aL2, Schwartz (2009: 66) makes the important distinction between linguistic development and ultimate attainment in child and adult L2ers. She argues that looking only at the end states of child and adult L2 acquisition is not sufficient because they may have reached the same end state via different routes. Instead, it is important to look at language development over time. Schwartz (2009) proposes the Domain by Age model that we shall discuss in some detail below.

To illustrate common developmental stages between cL2 and aL2, we shall take an example from Unsworth's (2005) work (as cited in Schwartz 2009). The Dutch language allows an operation called "scrambling," or movement of the definite object over negation, as (1) and (2) illustrate.

- (1) Base word order in Dutch, SOV
Nijntje gaat **niet de bloem** plukken.
Miffy goes not the flower pick
- (2) Scrambling of definite DO: DO Neg V
Nijntje gaat **de bloem niet** plukken.
Miffy goes the flower not pick
- (3) cL2 and aL2 order: (S Aux) Neg V O
*(Nijntje gaat) niet plukken de bloem.
Miffy goes not pick the flower
'Miffy is not going to pick the flower.'

When learning the possibility of the object scrambling over negation as in (2), both child and adult L2 learners go through a stage that is not attested in child L1 development. Under the influence of English, they produce sentences as in (3), in which the object appears after the verb and negation. Later on, they go through a stage when they produce both the attested and the unattested order, before they converge on the target word order as in (2). Children acquiring Dutch as a native language do not make such errors because they are not influenced by a previously learned language.

Unsworth (2008b) compares the scrambling behavior of low-proficiency, intermediate-proficiency, and high-proficiency children and adults L2 acquirers. The behavior that emerges is as follows: while the low proficiency children and adults fail to scramble where they should, intermediate and high-proficiency children and adults scramble more or less consistently.

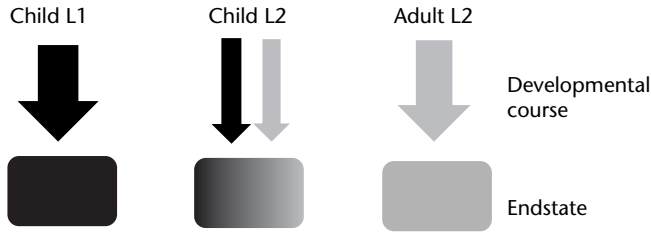


Figure 6.1 Illustration of the Domain by Age model by Schwartz (2009)

Thus, in this experiment, participants pattern together on the basis of proficiency, not on the basis of age. The L2 children and adults were observed to pass through the same developmental sequence, and hence their behavior was consistent with the claim that UG is involved in adult L2 acquisition.

Capitalizing on such findings, Schwartz proposes the Domain by Age model (Figure 6.1), essentially proposing that child L2 acquisition proceeds like adult L2 acquisition in the domain of syntax (dark grey (black) arrows), but it is like L1 acquisition in the domain of inflectional morphology (light grey arrows). The rationale for the second part of this claim came from work by Weerman (2002), which documents overgeneralization errors with Dutch adjectival morphology.² These overgeneralization errors go away by age 6 for Dutch L1 children, but they persist for child L2 learners.

However, more recently Schwartz (2009) brings forward unpublished data from Tran (2005), to update this latter part of the Domain by Age model. Tran investigated whether a well-supported fact in child German is also manifested in cL2 German. German children move the verb in second position in the sentence only when it is finite, that is, it is correctly marked with person, number, and tense inflectional morphology. When the verb is in its infinitival form, it takes sentence-final position in child speech. The data in (4) and (5) illustrating this contrast comes from Poeppel and Wexler's work on the production of Andreas, a monolingual German acquirer at age 2;1.

- (4) V2 (OVS) (Poeppel and Wexler 1993: 14, (13b))

Ein Fase hab ich
a vase have I

² L1 and L2 children produce a schwa in contexts where there should be no schwa.

- (5) V-final (SOV) (Poeppel and Wexler 1993: 11, (11))

Thorsten Caesar haben.

Thorsten C. (=doll) to.have

Tran tested native English children learning German in Honolulu. While the high-proficiency learners had already acquired the above contingency, the mid and low-proficiency children had not. They placed 57% of non-finite forms in the verb-second position in marked contrast to Andreas. Schwartz suggests that these data support the observation that cL2 learners may get to the desired targetlike state of knowledge, but they take different routes from L1 children.

Additional data in this respect come from Herschehnsohn, Stevenson, and Waltmunson (2005), who tested 6-year-old cL2 learners of Spanish in an immersion context, a year and a half after immersion started in kindergarten. The researchers looked at one inflectional morpheme—the marking of number in 3rd person subject agreement—and tested it by eliciting answers to questions. The children were only 38% accurate on the first testing occasion and 56% accurate on the second testing occasion, two months later. At the same time, it is well documented that child L1 learners of Spanish make very few commission or substitution errors³ with this inflectional morpheme. This study, again, supports the conclusion that cL2 development is not like cL1 development, even in the area of inflectional morphology. Another recent study on cL2, Li (2012), will be discussed in Exercise 6.4. While more data and further confirmation with fresh language pairs is always needed, the overall picture at this moment in time remains as in Figure 6.2 (see e.g., Blom 2008, Sopata 2010).

In summary, there is accumulating evidence that cL2s are like aL2 learners both in the area of syntax as well as in the area of inflectional morphology. They go through similar developmental stages and experience difficulties with inflectional morphology marking. A fundamental reason for this situation may be the presence of an already learned language in the mind/brain of the learners, that is, bilingualism itself. We shall discuss common characteristics of cL2, aL2, and heritage speakers in the last section of this chapter.

³ A commission error is when a speaker produces a morpheme where it is not needed, e.g., *You works*. An omission error is when a speaker does not produce a morpheme where it is needed, e.g., *He work*. A substitution error is producing one morpheme instead of another, e.g., *He working*.

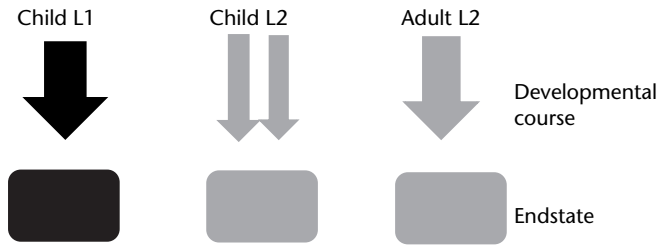


Figure 6.2 Illustration of the Domain by Age model, second version

6.3 L3/Ln acquisition

Although the focus of this text is on adult second language acquisition, a relatively new and vigorously developing area of investigation is third (L3) language acquisition, defined as the sequential acquisition of another language beyond a second language. In this section, when I write L3, I will actually include additional language (Ln) acquisition as well. How is L3 acquisition relevant to L2 acquisition? Studying multilinguals can actually help us to understand a lot about the linguistic representation of the second language in the mind. The essential issue, and one that is hotly debated among generative L3 acquisition researchers, is whether it is the grammar of the first or the second (or any subsequent) language that can influence the L3/Ln. The answer to this question is still related to the Critical Period Hypothesis, although in a slightly roundabout way. Here is the rationale. If the first language and the second language are equally capable or influencing the acquisition of subsequent languages, then they must have equal cognitive and epistemic status in the mind of the learner. If the learner is using her *entire* linguistic repertoire of grammatical features, constructions, functional morphemes, etc., then the linguistic knowledge of the second language must be represented in the mind in a way that makes subsequent transfer possible. If, on the other hand, the native and the second languages are learned in a qualitatively different way and result in qualitatively different representations, we cannot expect the L2 to have but a superficial influence on the L3/Ln.

Researchers in this field ask whether and when “transfer” takes places in the L3 interlanguage development. Transfer is defined as linguistic knowledge that can be traced back to prior linguistic experience (see definition in Chapter 2). To take an example given in Rothman (2015), consider the case

of English native speakers who also speak Spanish as an L2 and are learning Portuguese as an L3. English marks gender only on pronouns (*he, she, it*), while in Spanish and Portuguese every single noun is specified as feminine or masculine. In addition, adjectives agree with nouns for gender and number. We call the latter “grammatical” gender. A learner of L3 Portuguese with L2 Spanish will already be sensitive to grammatical gender being marked on all nouns and to adjective–noun agreement, exhibiting “early knowledge of morphological and syntactic reflexes for grammatical gender features” (Rothman 2015: 182).

In L3 acquisition, researchers consider the potential sources of transfer from any of the previously known languages, both in the initial state and in subsequent development. The specific interplay between the L1, the L2, and the L3 parameter values is being investigated. Does transfer in L3 acquisition come exclusively from the L1 as in L2 acquisition, as Leung (2006) has proposed? Does it come exclusively from the L2 (Bardel and Falk 2007) because it was the grammar most recently learned? Can it come from both languages, as Flynn, Foley, and Vinnitskaya (2004) have argued, or from no prior experience (Håkansson et al. 2002)? All of these possible sources of transfer have been evaluated, and found support from (some) experimental data. Linguists have also discussed whether transfer can be only facilitative (beneficial), or it can also be harmful, so that the L3 property is not really acquired, or is acquired with great difficulty. In this respect, the Cumulative Enhancement model (Flynn et al. 2004), as the name itself suggests, argues that transfer from the L1 or the L2 can be only facilitative, while Slabakova and García Mayo (2015) show that transfer of a very frequent and salient property can stand in the way of complete L3 acquisition. Needless to say, language proficiency in the second language will always be an important factor for transfer: after all, a learner cannot transfer what she has not really acquired.

Another variable that appears to play a role in L3 acquisition is what is usually referred to as the typological proximity of the languages. Let’s take for example Spanish and Portuguese, which are closely related languages belonging to the Romance language family; they can be considered typologically closer to each other than to English, a Germanic language. The idea of the Typological Primacy model (Rothman 2011, 2015), based on Kellerman (1983), is that the learner will transfer properties from the grammar, be it L1 or L2, which she perceives to be typologically closer to the L3. This typological relation may only be perceived, not real.

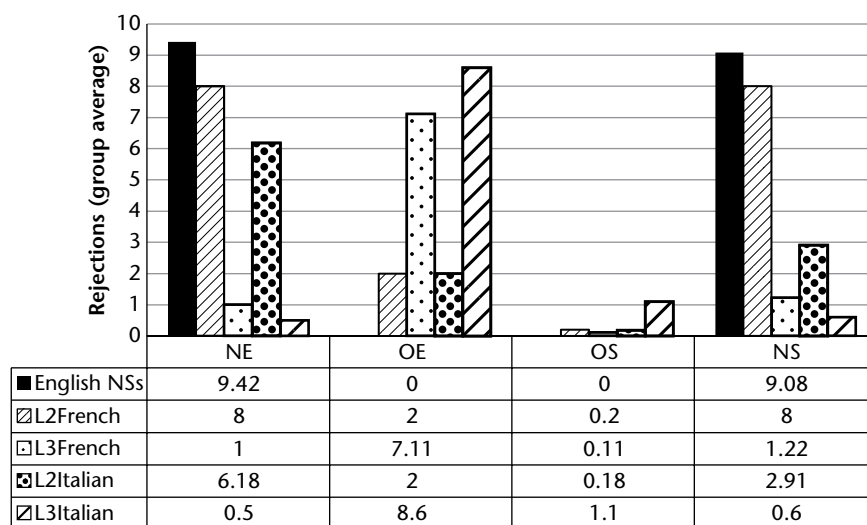
Finally, another issue debated in generative L3 acquisition research is whether transfer happens “wholesale” at the initial state, or whether it happens property by property and even feature by feature, from the initial state and into the development process. The Typological Primacy model (Rothman 2015) argues for the former view, while the Scalpel model (Slabakova 2015), defends the latter view. At present, there is no clear evidence that can answer this question definitively. Future research aiming to tease these positions apart should involve a longitudinal study of a third language, studied in adulthood *ab initio*, where the L2 and the L3, or the L1 and the L3, are typologically similar.

With so many variables to control, research on L3 acquisition involves complicated research designs. I shall look at three studies to give the reader a taste of the variety of language combinations studied. Flynn et al. (2004) is a pioneering study, one of the first to demonstrate empirically that the L1 is not the only source for L3 transfer at the level of formal syntactic features and functional categories. The authors examined the production of restrictive relative clauses in L1 Kazakh/L2 Russian/L3 English speakers. Relative clauses, as we have seen in Chapter 2, can have their head on the right or on the left of the clause. Head directionality in Kazakh is the same as in Japanese, while Russian and English work in similar ways. The researchers surmised that, if transfer is always from the native language, L3 acquisition of English by L1 speakers of Kazakh should resemble L2 acquisition of English by L1 speakers of Japanese (there is ample literature on this acquisition for comparison). However, the L1 Kazakh learners had an easier time acquiring the head directionality of the English relative clauses. Flynn et al. proposed that the L2 Russian had a facilitative effect on the acquisition of this particular L3 construction. More generally speaking, Cumulative Enhancement argues that experience in any previously acquired language can be taken advantage of in the acquisition of any subsequent language.

Rothman and Cabrelli Amaro (2010) investigated two L3s, French and Italian, where English was always the L1 and successfully acquired Spanish was always the L2. They examined properties related to the Null Subject Parameter. French is like English in this respect, while Spanish and Italian work similarly, that is, they are null-subject languages. For easier reference, the configuration is given in (6), where the italicized languages work the same way and the non-italicized languages work the other way.

- (6)
- L1 English* – *L2 Spanish* $\swarrow \searrow$ *L3 French*
L3 Italian

In addition, two groups of learners of French and Italian as second languages were tested. For this part of the experiment, five groups of participants judged the grammaticality (acceptability) of ten sentences in four conditions: null and overt expletive subjects (*It rains*) and null and overt pronominal subjects (*He snores*). The authors demonstrated that both groups of L3 learners transferred from Spanish, which is actually the wrong choice for L3 French. Even though English, the native language of the learners, could have helped in the acquisition of French, the learners were influenced by the typologically related Romance language. This finding is especially clear when we compare the learners of French as an L2 and as an L3, in Figure 6.3. These experimental results showed that transfer can be harmful, including when the transfer is from the language perceived to be typologically closer to the L3. In other words, they found support for the Typological Primacy model.



Notes: NE = null expletive subject; OE = overt expletive subject; OS = overt pronominal subject; NS = null pronominal subject

Figure 6.3 Group average rejection of null and overt subjects in a grammaticality judgment task, from Rothman and Cabrelli Amaro (2010)

Before going forward, let us spend some time understanding the figure. What is plotted, here, acceptance or rejection? What is the highest score each group of participants could have achieved? Each column stands for a score of one participant group in one condition; conditions are grouped together. Look at the first column in each group. Do the English native speakers behave predictably? Now look at the next two columns in each group. These are average scores of learners of French. Are their grammaticality judgments of null and overt pronominal and expletive subjects different? What can you say about the learners of Italian? The crucial groups of columns to notice are the first and the last ones, where the L3 French learners do not reject null expletive and pronominal subjects as they should, under the influence of their L2 Spanish.

Finally, another recent study demonstrated non-facilitative transfer, too, but from a first as well as a second language. Slabakova and Garcia Mayo (2015) investigated knowledge of English topicalization as in (7), a construction which moves the object to the front of the sentence when it is known, or already mentioned.

- (7) A: Did Janice like the wine?
 B. Oh, **the wine** she didn't drink (*it). She stuck to lemon ices.
- (8) El libro lo compr-amos ayer.
 the book ACC.CL.3M.SG buy-PAST.1.PL yesterday
 'The book, we bought yesterday.'

The languages under investigation were Basque and Spanish as L1 or L2, English was the third language, as (9) illustrates. Again, the presence/absence of italicization indicates similar constructions. As in Rothman and Cabrelli Amaro, there was a group of native Spanish speakers learning English.

- (9) **L1 Basque – L2 *Spanish* – L3 English**
***L1 Spanish* – L2 Basque – L3 English**

While English and Basque work similarly with respect to topicalization, the Spanish construction⁴ in (8) requires a clitic pronoun *lo* to double the moved

⁴ Known as "clitic left dislocation."

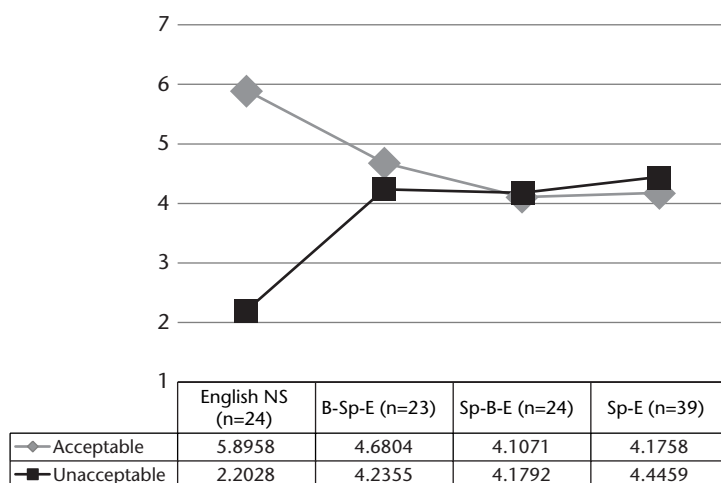


Figure 6.4 Mean ratings of topicalization by four groups of participants, from Slabakova and García Mayo (2015)

object. Therefore, the unacceptable test sentences in English had a pronoun in the object position, as in (7). The experimental design involved listening to recorded conversations and evaluating the acceptability of test sentences with and without resumptive pronouns, on a scale of 1 to 7. Figure 6.4 plots the mean ratings of all the participant groups.

Look at Figure 6.4 and explain what you see. Do the native speakers exemplify a contrast between acceptable and unacceptable topicalization? What do mean ratings of around 4 signify in the learner groups' performance?

Slabakova and García Mayo argued that non-facilitative transfer from Spanish is exhibited in their results, no matter whether Spanish was the first or the second language of the trilinguals. The bilingual results also buttress this conclusion.

Teaching relevance: "Concealed" L3 learners

It is very interesting that after researchers started paying attention to L3/Ln acquisition, it turned out to be the case that many classroom learners are in fact not L2 but L3 learners. For example, it is most often the case in the US college system that students come to Portuguese or French or Italian classes with some or a lot of Spanish L2. It is useful for teachers of such students to be aware of how the Spanish grammar can facilitate these other Romance languages as L3s.

In summary, the vigorously developing research in L3/Ln acquisition focuses on the issue of the source of cross-linguistic influence. Transfer can come from the first or the second language of the learners, and it can be facilitative or harmful. As in generative linguistic research in general, transfer is understood not as a superficial phenomenon of influence, but on a deeper level of grammatical competence and development. Generative language acquisition researchers do not perceive L3/Ln acquisition as a simple extension of L2 acquisition, because of the complex sources of transfer available to multilinguals.

6.4 Heritage language learners

We brought forward the importance of heritage language speakers for the Critical Period debate in Chapter 4. In this section, we will expand on the characteristics of this important population's language knowledge.

The term *heritage speaker* was first introduced in the mid-1970s (Cummins 2005) but has been gaining ground in the acquisition literature since the 1990s. Broadly defined, heritage speakers are child and adult members of a linguistic minority who grew up exposed to both their home language and the majority language. For some researchers, this definition also includes indigenous languages, not just immigrant languages (Fishman 2006). Spanish, South Asian, Russian, East Asian, and Arabic speakers in the US who are the so-called second generation immigrants are representative examples. While the parents are either monolingual or dominant in their native language, the children grow up in homes where both the majority language and the native language are spoken. Very commonly, the heritage language is the individual's native, or consecutively first language, learned in a naturalistic setting in a family environment. The heritage language could also be one of two simultaneously acquired L1s. However, the dominant language of those individuals is the majority language of the country and the community, exerting educational and social pressures over the weaker, heritage language. Once the majority language is introduced, the heritage language becomes restricted to family-centered uses and situations. Figure 6.5 and 6.6 from Montrul (2012) illustrate the typical development of heritage and L2 languages, where the length of the columns is meant to represent proficiency with respect to adult norms.

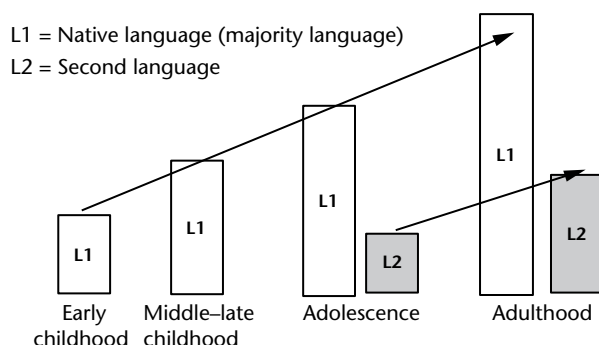


Figure 6.5 Typical development of a first (L1) and second (L2) language (after puberty) in a majority language context, after Montrul (2012)

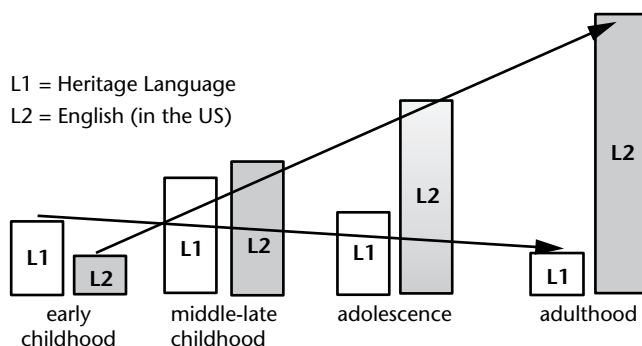


Figure 6.6 Typical development of a heritage language (L1) in a majority language context, after Montrul (2012)

Who exactly falls into this rather heterogeneous group of heritage speakers? According to Montrul (2011b), this group may include:

(a) simultaneous bilinguals, those exposed to the heritage and the majority language before the age of 3–4; (b) sequential bilinguals or child L2 learners, those exposed to the heritage language at home until age 4–5 and to the majority language once they start preschool; and (c) late child L2 learners, children monolingual in the heritage language, who received some elementary schooling in their home country and immigrated around ages 7–8. (Montrul (2011b: 157)).

This is a good place to reinforce an important distinction in bilinguals and multilinguals. One should be careful not to confuse *language dominance* with *language chronology* or *nativeness*. Heritage speakers have acquired their heritage language chronologically first and from birth. It is their

mother tongue. However, they later become dominant in another language, spoken by the wider community around them. Very often they sound nativelike in that second language. Countries with consistent streams of immigration through the years, such as Canada, the UK, and the US are full of such people, the so-called second-generation immigrants (Silva-Corvalán 1994). Teachers of foreign languages in such countries encounter many individuals fitting this profile in their classrooms. When tested in their weaker, non-dominant language, these speakers may be more or less proficient. Indeed, according to some sources, about 30% of undergraduate students in American colleges and universities may be heritage speakers of an immigrant language; in California, this percentage may be even higher, at around 40% (Carreira and Kagan 2011).

Teaching relevance: Dominance versus chronology

Identifying specific areas of linguistic knowledge in which the heritage language speakers and L2 learners may or may not differ informs materials development, classroom-based instructional intervention, or language program direction. The educational needs of these two populations are just not the same. It is important for teachers to be aware of just how different heritage speakers are from adult L2 learners.

Which aspects of language suffer when the consecutively first language gets used less and less over the years, until it becomes a weaker language of a bilingual? Recent research into the competence of these speakers has uncovered significant gaps in their linguistic competence, making them very different from age-matched monolingual speakers.⁵ In general, adult heritage speakers are better at aural comprehension than oral production. This is not surprising if we keep in mind that they continue to overhear the heritage language in their family circles although they may not speak it on a regular basis. Furthermore, their written language skills are worse, compared to adult L2 learners, because they typically do not have the benefit of schooling in the heritage language.

⁵ See Benmamoun, Montrul and Polinsky (2013a,b) for a recent overview and linguistic treatment.

An important early series of studies, Au et al. (2002) and Knightly et al. (2003), compared beginning L2 learners of Spanish to Spanish native speakers with only a receptive knowledge of the language. They labelled the latter speakers “overhearers.” The researchers found that the heritage speakers were significantly more nativelike on the phonetics/phonology and pronunciation measures, while their performance on morphosyntactic measures, around 60% accurate, was similar to the L2 performance. Advantages of heritage speakers on phonetic discrimination and pronunciation have been confirmed for various heritage languages: Saadah (2011) for Arabic, Lukyanchenko and Gor (2011) for Russian, Chang et al. (2008) for Mandarin Chinese.

Although heritage speakers enjoy some advantages in phonetics/phonology, their linguistic systems share many similarities with adult L2 speakers. Many of the problem areas typical of L2 learners, such as inflectional morphology, complex syntax, syntax–semantics mismatches, and discourse-related meanings seem to be problematic for heritage speakers as well. We will illustrate the parallels between heritage and adult L2 learner grammars with some concrete studies below.

One representative study of heritage speaker case morphology, Song, O’Grady, Cho, and Lee (1997), investigated the ability of monolingual Korean and heritage Korean children to use case marking for distinguishing the doer of the action in a transitive situation. In Korean, the order of the object and the subject can be reversed without any additional changes, because the arguments are marked with Nominative and Accusative case. Participants heard a scrambled Korean sentence as in (10) (similar to the scrambling in Dutch we saw earlier). They had to choose one of two transitive construals (who is hugging who), as illustrated in Figure 6.7.

- (10) oli-lul thokki-ka anacwue OSV
 duck-ACC rabbit-NOM hug
 ‘It is the duck that the rabbit is hugging.’

Whereas the monolingual children responded above chance (50%) by age 4, even the 8-year-old heritage children did not reach chance level on these structures. Since Korean case marking depends on four interrelated factors—grammatical function, focus, animacy, and definiteness—the researchers concluded that the input the heritage speakers received was

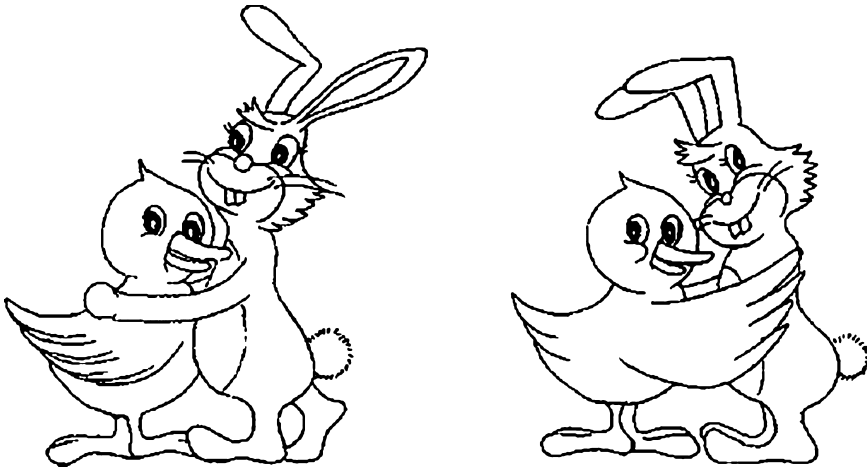


Figure 6.7 Sample picture used to test the attentiveness to case in the interpretation of sentence (10) by Korean monolingual and heritage children, from O'Grady et al. (2011) Reproduced with permission

not sufficient for them to establish a difficult form-to-meaning mapping of this sort.

In another recent study, Albirini, Benmamoun, and Saadah (2011) investigated the narrative production of Arabic-speaking heritage speakers of the Egyptian and Palestinian dialects. They attested quite a high level of proficiency among these speakers. Still, some significant gaps in their knowledge became visible, in comparison to monolinguals. They had problems with agreement and tense morphology. For example, where inflected verbal forms were expected in the narratives, these speakers frequently substituted a simpler participial form. The researchers contend that the heritage speakers use this participial form as default, when access to the verbal paradigm fails them. Transfer from English word order was also attested.

To exemplify how heritage speakers deal with complex syntax, we shall mention the findings of Polinsky (2011) on comprehension of relative clauses in heritage Russian. Polinsky tested prepubescent heritage speakers, age-matched monolingual Russian children, and adult Russian speakers. The child heritage speakers performed on a par with the monolingual children. Meanwhile, the adult heritage speakers had significant problems with relative clauses. A very interesting feature of their linguistic competence was uncovered: they performed at chance on object relative clauses (*The dog that the girl saw* ____) but were close to the other experimental

groups in their comprehension of subject relatives (*The girl who _____ saw the dog*). Polinsky argued that the nativelike grammatical knowledge of relative clauses that the child heritage speakers showed had been reanalyzed by the adult heritage speakers into a new system allowing only subjects, but not objects to be heads of relative clauses.

The study of heritage speakers' competence, just as the study of cL2A, has been developing very vigorously in the last ten years. Three major explanations have been proposed for the linguistic development of heritage speakers. Of course, all such explanations have to refer to differences between heritage speakers and monolinguals, as well as to the similarities between heritage speakers and adult L2 learners. One line of explanation proposed by Montrul argues that heritage speakers exhibit signs of *incomplete development* (Montrul 2008). The reduced situational, family-based usage can cause "arrested development" in the heritage language or lead to incomplete acquisition of some constructions. Another explanation, proposed by Polinsky maintains that heritage languages have in fact been acquired completely, but have subsequently reduced by *language attrition* (loss of native language, see the following section) (Polinsky 2011, among others). A third explanation, argued for by Sorace and Rothman separately, directly relates heritage language competence to reduced input, not only quantitatively but qualitatively. Heritage speakers receive input primarily from speakers—first-generation immigrants and other second-generation speakers—who are themselves living in a language contact situation and hence may be consistently providing them with *linguistically changed input*. Thus, individual attrition may be reinforced through intergenerational attrition (Sorace 2004, 2012, Rothman 2007, Pascual y Cabo and Rothman 2012). Of course, it is likely that all of these factors work in consort to some extent. The common thread running through these three explanations is that the linguistic input, variable both in quantity and in quality, and the reduced language use, are the chief reasons why heritage grammars are different from monolingual grammars. This is true even though heritage language speakers are native speakers.

6.5 Language attrition

The term *language attrition* refers to native language regression or loss. The speakers of such a reduced language are dubbed *attrited speakers*. Attrition

research examines the potential erosion of native language competence after long exposure to another language. Although they are related, heritage language competence and language attrition have to be clearly delineated. In the case of attrition, a given property of language has been fully acquired by a speaker who has typically reached a stable native grammar state, but is subsequently lost due to reduced exposure to the native language. Attrition is much easier to demonstrate in adults than in children: the latter may or may not have acquired the linguistic property completely in the first place. Recall from the previous section that Polinsky (2011) argued for attrition being the cause of heritage speaker differences by comparing heritage children who exhibited knowledge of object relatives with heritage adults, who didn't. Only this type of research design can rule out incomplete acquisition and support attrition as the sole cause of the grammatical deviation from native norms.

In cases of native language attrition, exposure to an L2 is often sustained and intense and may be accompanied by a sharp decrease in exposure to the L1. The typical attrited speaker is the first-generation immigrant, who may be the parent of heritage speakers. The definition of attrited speakers, similarly to heritage language, encompasses attriters with a wide variety of linguistic situations and profiles (Schmid 2011). Under a broad definition, attrition can be studied in balanced simultaneous bilingual or childhood L2 learners, adults, as well as heritage speaker bilinguals. A narrower definition focuses on loss of (some parts of) the language faculty after the putative critical period.

Language attrition is typically minimal if attriters lost contact with their native language in adulthood. For example, Schmid (2002) found that, after more than 30 years of living in the US, German Jews exhibited some transfer from English but made very few morphosyntactic errors that could be attributed to L1 attrition. No adult speaker, who comes into contact with a second language and starts living in a new environment, is likely to forget the verb conjugations, the native sound contrasts, and how to ask questions in their native language (Keijzer 2007). But less dramatic, although still measurable, changes in native grammars may be documented when potential attriters are compared to recent arrivals or native speakers in the country of origin (Gürel and Yılmaz 2013, Sorace 2005).

However, not all cases of attrition are relatively mild and superficial. Iverson (2012) discusses a case study of extreme attrition of native Spanish in contact with Brazilian Portuguese. The bilingual speaker, called Pablo

(not his real name), was tested in his fifties. He was born in Chile, where he went to school until age 13, then worked in Chile and in other Spanish-speaking countries until he moved to Brazil in his early twenties. At the time of the testing, Pablo had spent 30 years in Brazil with minimal contact with Spanish. He never visited Chile nor did he speak to family members there on a regular basis. He is a street artist, married to a Brazilian Portuguese-speaking woman and is the father of a Brazilian Portuguese-speaking daughter. He rarely encounters other Spanish speakers.

Iverson obtained spontaneous speech samples which demonstrate that Pablo intersperses many Brazilian words in his Spanish narrative, not only nouns, but verbs as well. Sometimes he produces words that are neither Spanish nor Brazilian Portuguese but amalgamations of both, such as *enton* 'then' from the Brazilian *então* and the Spanish *entonces*. His speech, which he considers Chilean Spanish, is not recognized as such by native speakers. Iverson further tested Pablo, as well as two control groups of monolingual Brazilian Portuguese and Chilean Spanish speakers of the same socio-economic status as Pablo, and speaking the same dialects. A range of tasks probed judgments of verb–subject word order, overt pronoun use in discourse-neutral contexts, null objects in complex syntactic contexts and/or with definite antecedents, interpretation of embedded overt subject pronouns in ambiguous contexts, and relative clause attachment in ambiguous contexts. Although Spanish and Brazilian Portuguese are closely related Romance languages, the tested areas represent contrasts between those grammatical systems. The findings were extremely robust: Pablo's performance diverged dramatically from the Spanish control group on the whole range of the tested constructions. He consistently performed qualitatively (and often quantitatively) like the Brazilian Portuguese control group. Iverson's conclusion is that the whole of Pablo's native Spanish grammar may have restructured under the influence of the second language grammar in the circumstances of extreme cessation of native input and usage.

In summary, we speak of language attrition to describe changes in the native grammar when immigrants stop using their native language and carry out most of their communication in the second language, the language of their new linguistic community. There could be personal and societal pressures for such disuse. Research to date has shown that the changes to the native grammar, in most cases, remain superficial and are manifested in loss of processing accuracy and speed, as well as labored access to lexical items and complex constructions. However, extreme cases such as Pablo's are also

attested. Future research should address the factors making possible and accelerating such restructuring of a native grammar through disuse.

6.6 Commonalities and differences between the four acquisition contexts

In Chapters 5 and 6, we have been describing various types of bilingual situations, as well as the effects of the first language on the second and the effects of the second language on the first. In Chapter 5 we discussed bilinguals who have two first languages (2L1), or simultaneous bilinguals. In Chapter 6 we discussed child second language (cL2) and adult second language (aL2) learners, multilinguals, heritage native speakers, and attrited native speakers. Now is a good time to take stock and identify which are the central and decisive factors affecting all of these language development processes. We shall consider Age and Input as crucial impacts, admittedly simplifying matters and summarizing a lot of studies for the sake of gaining some overarching insight into bilingual language use and maintenance.

6.6.1 Is age the crucial factor in bilingual acquisition?

The short answer is no. Evidence supporting this negative answer comes from three relatively well-established facts. One fact is that even if a child starts acquiring two languages from birth, one of them is likely to be the weaker language as a function of relative hours of exposure and hours of use. There are many indications that the mere fact of becoming a (proficient) bilingual has ramifications for both linguistic systems. For example, Flege (1987) demonstrated that bilinguals pronounce some phonemes with some intermediate value in both their languages, a value that is not attested in either language. Dussias (2004) found a similar bidirectional interference effect with respect to grammatical processing strategies. Such findings suggest that the monolingual norm may be something that bilinguals can never fully attain, not necessarily because they have reached the limit of their acquisitional potential but simply by virtue of being bilingual.

A second fact militating for a negative answer to Age as the crucial factor is that child and adult native speakers follow the same developmental paths and make similar developmental errors. As Schwartz and others have argued, if Age were the crucial factor, we would expect to see children and

adults diverge in the stages of acquisition they exhibit and the type and rough percentages of errors they make, but they don't.

A third factor is possible ultimate attainment in some areas of the grammar, even if one starts in adulthood. Although it is true that not all bilinguals become near-native speakers, it is possible for many to become such speakers in the areas of syntax, semantics, or pragmatics. The level of proficiency attained is usually a better predictor of truly nativelike behavior than age of acquisition (Wartenburger et al. 2003). Although age and proficiency may be very difficult to disentangle, and they are seldom controlled for, there are studies which point to the possibility for highly proficient L2 speakers who started learning the language across a range of ages to apply the same processing mechanisms as natives (Herschensohn 2007). To give just one example among many, Montrul and Slabakova (2003) investigated knowledge of aspectual semantics in very advanced-proficiency bilinguals and identified around 30% of them who performed within native speaker norms. If age is not crucial for the successful acquisition of such properties, then the critical period claim is too strong indeed.

6.6.2 The Critical Period Hypothesis and the importance of the input

If age is not the absolutely crucial factor, then what is? Some of the points discussed in the previous sections indicate that actually input is the crucial factor. To elaborate, simultaneous bilinguals are exposed to two languages from birth, but the effects of bilingualism are mitigated by the amount of time they spend using the weaker and the more dominant language. Heritage speakers are also exposed to the family language from birth, but the disuse of that language in later years in favor of the majority societal language leads to dramatic reduction of fluency and accuracy, and in many cases even to restructuring of the native grammar. Finally, attrited speakers who left their native country in adulthood, and thus interrupted their native language flow of communication, also lose some fluency and experience lexical and constructional access difficulties. In extreme cases, they may undergo changes in underlying mental representations, not just processing preferences. The common thread between these three different populations: simultaneous bilinguals, heritage speakers, attriters, is the loss of usage and reduced availability of native language input. When input is lost or diminished, there are consequences for the nativeness of the grammar.

6.6.3 What kind of input?

Is any input going to be sufficient for language maintenance? The short answer is that, for a language acquired to completeness to be maintained, the input has to be copious, sustained, and continuous. Communication (give and take) in that language is essential. Communication has to be on diverse topics, allowing the language users to comprehend and produce a variety of constructions of different complexity. Reading in the language, not just oral communication, is also considered beneficial for language maintenance, because there are different constructions and aspects of language exemplified in oral and written varieties. For example, it is a well-known fact (and easily checked) that heritage speakers tend to be good at aspects of language that are reinforced in family communication, such as household, food, and vocabulary related to everyday life. They have been exposed to the lower frequency, academic-oriented vocabulary, and the complex structures of the more formal registers in their second language in school settings, and they may not have learned that more specialized vocabulary or registers in their native language.

6.6.4 Which areas of the grammar suffer with reduced input?

The answer to this question comes from looking at the performance of bilingual speakers in reduced input conditions on a variety of linguistic properties. Benmamoun, Montrul, and Polinsky (2013a,b) cite morphology, complex syntax, semantics, and discourse pragmatics as areas of the grammar where heritage speakers display non nativelike characteristics. For example, Montrul and Ionin (2010) identified non targetlike interpretations of the definite article and the bare noun in generic contexts. Heritage speakers accepted sentences such as **Leones son peligrosos*, ‘Lions are dangerous,’ which is good in English but unacceptable in Spanish. Heritage speakers have been shown to omit case marking or misuse it (Montrul, Bhatt, and Bhatia 2012, O’Grady et al. 2011). Polinsky (2006) showed that Russian heritage speakers who cannot write in Russian had reanalyzed the gender system to contain two genders instead of the monolingual Russian three. Kim, Montrul, and Yoon (2009) studied the interpretation of three local and long-distance reflexives by Korean heritage speakers. They found that while the monolinguals had

a three-way anaphor system, the Korean heritage speakers had a simplified system with only two pronouns.

In sum, studies have uncovered morphological instability, high variability, and nonnative knowledge of definiteness marking, gender agreement, case marking, lexical aspect, grammatical aspect, mood, and inflected infinitives. As this already sizable body of data indicates, native speakers in circumstances of reduced input demonstrate nonconvergence, simplification, and/or reanalysis of various areas of their native grammar. Frequently, these have been shown to be the effect of the dominant second language. It appears that all areas of the grammar can be affected by reduced input, with the possible exception of phonetics/phonology. However, it is still an unanswered question why some properties within an area of the grammar are affected while others are not. For example, Håkansson (1995) found that Swedish heritage speakers did not have problems with the Verb Second rule, a complex grammatical rule. Furthermore, while Russian heritage speakers have severe problems with gender and case, Spanish heritage speakers typically have fewer problems with these grammatical morphemes. Theoretical answers to these questions should include an array of experience factors affecting linguistic knowledge, as well as principled proposals based on linguistic theory.

6.6.5 Are first and second language acquisition qualitatively different?

The short answer is no, first and second language acquisition are not qualitatively different. If the Language Acquisition Device were permanently altered after the acquisition of the native language in such a way that subsequent languages were acquired in a qualitatively different way, then the acquisition of the dominant language of heritage speakers cannot be explained adequately (see Figures 6.3 and 6.4). Furthermore, if we find, as we do, that even adult L2 learners attain similar proficiency to monolinguals in some areas of the grammar, then the same language acquisition mechanisms must be at work. It is as simple as that: if language acquisition processes (L1A, L2A, L3/LnA, heritage L1A, heritage L2A) start at different ages but lead to comparable attainment levels, the mechanisms cannot be qualitatively different.

This interpretation of the L1–L2A fundamental difference remains controversial. Proponents of a strong version of the Critical Period Hypothesis argue that it is a special cognitive-based and measurable aptitude that

makes some but not all individuals better language learners, leading to performances quantitatively comparable to monolinguals (Long, DeKeyser, Meisel). According to this line of reasoning, even if they are better language learners, these successful individuals do not learn language in the way L1 children do, because they rely on general cognitive skills, as well as metalinguistic patterns of observation and explicit explanation. This type of argument is only countered by demonstrating that successful L2 learners have mastered properties that are not taught in language classrooms and that are not easy to acquire based on observation of natural language input. Such research findings exist, and we will see them discussed at length in Chapter 10. Suffice it to mention just one study here, Slabakova (2003), which shows that successful interpretation of taught and untaught semantic contrasts is demonstrated by Bulgarian learners of English even at intermediate levels of proficiency.

At the same time, if we maintain that L1 and L2 acquisition are fundamentally similar processes, we must explain the wide discrepancy in attainment levels for the two types of acquisition. While a typically developing first language is always acquired to completeness, a typically developing second language is not *guaranteed* 100% success. Could the solution lie in the fact that acquiring a second language creates a bilingual individual? Bilinguals experience complex demands on their language faculty because they must constantly negotiate two competing systems at the same time, both in production and comprehension. Even if they are able to establish fully targetlike underlying knowledge of the L2 grammar, they may fail to apply it deterministically due to the competing linguistic systems and the heightened demand on processing resources. In order to use her weaker language, a bilingual has to expend a great deal of effort at inhibiting the deeply entrenched routines used by her dominant and stronger language. This inhibition can sometimes fail, allowing the output from underlyingly intact rules to show influence from the dominant language.

It stands to reason that all types of bilinguals (consecutive, sequential, and adult) should show evidence of such “bilingualism effects.” In other words, we should find more widely variable outcomes with all types of acquisition leading to bilingualism but no such variation with monolingual acquisition (barring impairments and traumas, of course). Interestingly, such evidence comes from the area of simultaneous bilingualism. “Raising children to speak a single language has a 100% success rate except in some cases of impairment. Raising children to speak two languages only has a

75% success rate.” (De Houwer, 2007: 421). Why would that be the case, even with consistent exposure to two languages from birth? We can also construe the heritage language findings discussed in this chapter in this light: while competence in one language goes up, competence in the other language might go down, see Figure 6.6.⁶

A clinching argument comes from a recent study by Hopp and Schmid (2013). The idea was to compare the performance of late L2 learners against that of other, similarly fluent and proficient bilinguals, who share the effects of cross-linguistic influence but who have acquired the same language from birth. The two researchers compared data that they had collected separately⁷ on some comparable measures. Hopp’s body of data was from near-native speakers of German with English and Dutch as their native languages; Schmid’s was from German attriters in Dutch and English environments. Since the two populations took a C-test⁸ measuring proficiency, their global proficiency could be compared directly. The ranges of the attriters on the C-test were much larger, but the means were comparable, and lower than the monolingual control group.

An investigation of perceived foreign accent among these groups of speakers revealed that a minority of the L2ers (37.5%) and a majority of the L1 attriters (72.5%) were perceived to be within the monolingual range. However, 80% of all L2ers fell into the range delimited by the attriters. Schmid (2013) further compares the rate of morphosyntactic errors per minute, taken from a spoken speech sample. Although the topics and the duration of the samples were not exactly the same, Schmid collated errors in four categories: lexical errors, word order (syntax) errors, and two morphological errors (case and gender) (see Figure 6.8).

Let’s examine the interesting contrasts between attriters and near-natives on the data in Figure 6.8. The near-native L2 speakers of German are less accurate on the lexicon (which Schmid suggests could be due to the specific

⁶ The reader should not extend these arguments to suggest that there is only room for one language in the brain and if you add another, the first will inevitably get worse. This is not the case. All I am suggesting here is that languages suffer from loss of input.

⁷ Hopp (2007) and Schmid (2007).

⁸ A C-test is a type of language test in which learners read a brief paragraph in the target language. The first two sentences are left intact. Thereafter, every other word is left intact but for each alternate word, only the first half of the word is written out while the second half is represented by a blank space for each missing letter. This is what it looks like: every oth_ _ word i_ left int _ _ _.

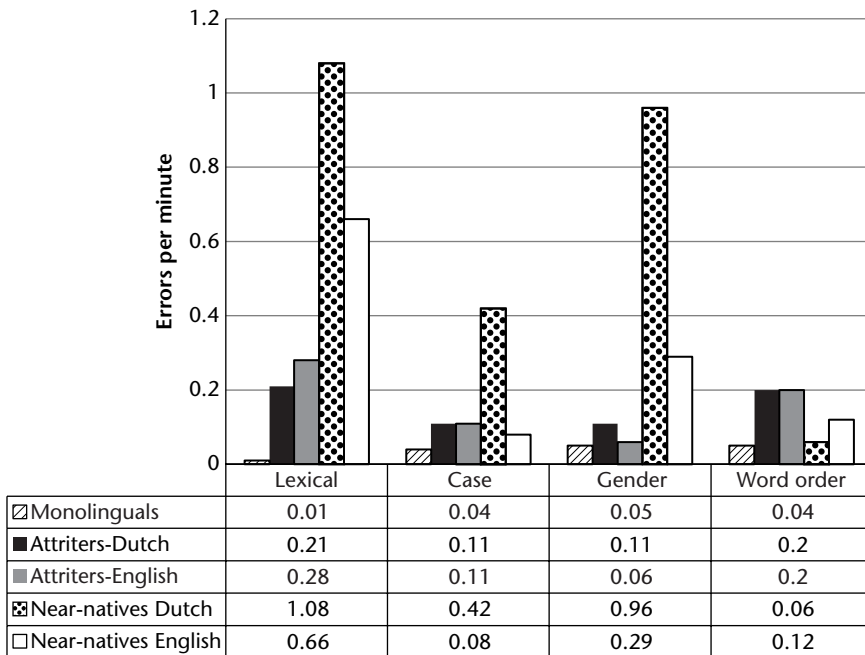


Figure 6.8 Errors per minute in the production of four experimental groups, from Schmid (2013)

speech topic), but they are extremely accurate on word order, even more so than the attriters. With respect to case marking, the English native speakers are again out performing the German attriters. Gender-marking, on the other hand, which is relatively unproblematic for the attriters, seems to be hard for the near-natives. This is particularly prominent for the Dutch-native learners of German, although Dutch has a system of gender-marking that is quite similar to German. We shall look at the acquisition of grammatical gender later (in Chapter 7), suffice it to say here that it is considered one of the hardest functional categories to acquire due to the high level of lexical learning involved. The overall results of this interesting and rare comparison suggest that it is *possible* for L2 learners to become as successful as bilingual natives on many linguistic measures, while monolinguals outperform both, at least at the group level. As the research design keeps constant the effect of managing two linguistic systems online, both the similarities and the differences between the two populations must be due to the underlying competence or the performance of the two groups.

In sum, the position emphasized in this chapter is that continued diverse and comprehensible linguistic input and rich communicative use emerge as the indispensable conditions for successful language acquisition, overriding potential critical periods (at least in some modules of the grammar). At the same time, Universal Grammar principles and parameters constrain learners' hypotheses and can facilitate subsequent language acquisition. Bilingualism or multilingualism effects, such as the necessity to manage two or more competing grammar systems in production and comprehension, may go a long way towards explaining the higher variability in additional language attainment (no matter whether that is from birth, in childhood, or in adulthood). It could be the case that the language aptitude some researchers correlate with attainment is directly related to the ability to navigate two competing grammatical systems, although this remains an empirical question. I have argued that the null hypothesis of L2 acquisition is that it is fundamentally similar to L1 acquisition,⁹ because both are processes of *human language acquisition*.

6.7 Exercises

Exercise 6.1. Montrul et al. (2008) revisited the question originally posed by Au et al. (2002): whether early exposure to the language confers an advantage in linguistic ability to heritage speakers ($n = 42$) over L2 learners ($n = 44$). This study investigated syntactic knowledge of gender agreement in three tasks: a comprehension task, a written morphology recognition task, and an oral production task. While the native speakers performed at ceiling on all tasks (with almost 100% accuracy), the two experimental groups were significantly less accurate. Yet, the results also revealed interesting *task effects*. Within the L2 group, speakers were significantly more accurate on the two written tasks ($M = 89.5$ and $M = 88.5$) than they were on the oral production task ($M = 72.1$). The heritage speakers, by contrast, were more accurate on the oral task ($M = 89.7$) than on the two written tasks ($M = 84.6$ and $M = 83.3$). Oral versus written accuracies are significantly different within groups. Furthermore, comparing between groups, L2 learners were more accurate than heritage speakers on the two written tasks,

⁹ See also Hopp (2007).

while the heritage speakers were more accurate on the oral task than the L2 learners.

Discuss whether these task effects change substantially, or change at all, the interpretation that we have espoused in this chapter: that heritage speakers' competence is comparable to L2 competence, due to their reduced language input and language use.

Exercise 6.2. Find the study from Exercise 6.1 in your local library or online, read it and discuss whether the interpretation of the authors coincides with the interpretation stated above. Discuss whether written or oral tasks better capture implicit linguistic competence. What do the overall results tell us about the linguistic competence of these two groups on the property under investigation?

Exercise 6.3. Still on the same study, consider the Figures 6.2, 6.4, and 6.7. In which group is there more variability? How is that captured in the graphs? Now look at Table 10. How were the individual results calculated? Discuss whether the individual results support the group findings and your comments on variability within groups.

Exercise 6.4. A recent longitudinal study of the initial state in child L2 acquisition, Li (2012) studies six 7- to 9-year-old Chinese-speaking children learning English in an immersion context in the US. The children had resided in the US between four and five months at the beginning of the study. The school instruction and environment was in English only. The children's spontaneous productions were recorded once a month, for eight months in interactions with the researcher. There was not much development in the children's data over that time. The findings of the study are summarized in Table 6.1. Note that commission errors (producing a wrong morpheme) on tense and agreement are around 13%.

Lardiere (1998) describes the production of the adult L2 learner Patty. Find the article, print it out, and read it. Answer the following questions:

Table 6.1 Percentage of correct usage in the spontaneous production of six English-learning children

Correct use of verb inflections				Syntactic properties	
3PSG -s	Regular -ed	Irregular past	Copula be	Overt Subjects	Nominative Subjects
16%	13%	38%	93%	~100%	~100%

Question 1: Compare the personal characteristics of Patty and the children in Li's study. Then compare the rates of correct suppliance for Patty and the children on the properties listed in Table 6.1. Make a combined table. Do you see a similar pattern?

Question 2: How does Lardiere explain the discrepancy between the accuracy reflected in the first three columns in Table 6.1, and in the last two columns? Do you think that Lardiere's explanation of Patty's performance can extend to the children's performance?

Question 3: Lardiere describes Patty as a fossilized learner. Why? Do you think that the children in Li's study are fossilized learners, and why?

Exercise 6.5. In Section 6.3, we discussed various models attempting to describe the L3/Ln acquisition process. Looking at the experimental designs described in this section, provide experimental configurations of languages and properties to test the following hypotheses:

Hypothesis A: Transfer can come only from the L1;

Hypothesis B: Transfer can come only from the L2.

What findings would support hypothesis A? What findings would support hypothesis B?